MILITARY SPECIFICATION

CONTROL GROUP, ELECTRIC POWER OK-XXX(V)/A
GENERAL SPECIFICATION FOR

This specification has been approved by the Naval Air Systems Command, Department of the Navy.

1.0 SCOPE

1.1 Scope. The equipment covered by this specification shall perform the management function for aircraft electrical systems. The management function includes data multiplexing, demultiplexing, processing, and built-in-test. The control group accepts discrete control and indication inputs from bilevel signal sources, multiplexes and transmits these signals to the data processor. The input data samples are demultiplexed and used as terms in the power management equations which are solved by the data processor. The equation solutions are multiplexed, transmitted, demultiplexed and translated into continuous bilevel control signals to power switching or indication devices.

1.2 Classification. The Control Group covered by this specification (Figure 1) is composed of:

Type Designation	Number of Units	Applicable Para.
Data Processor(s)	2 maximum	3.5.3
Crew Control Panel	1	3.5.4
Multiplexers (64 discrete inputs)	32 maximum (each individually addressable)	3.5.5
Demultiplexers (64 discrete outputs)	32 maximum (each individually addressable)	3.5.6
Multiplexers/Demultiplexer (64 inputs/64 outputs)	32 maximum (each individually addressable)	3.5.7
Data Link Coupler/Terminator	72 maximum	3.5.8

Multiplexer, demultiplexer, multiplexer/demultiplexer, Crew Control Panel (Multiplexer/demultiplexer) terminals may be intermixed up to a total of 32 multiplexer and 32 demultiplexer functional addresses.

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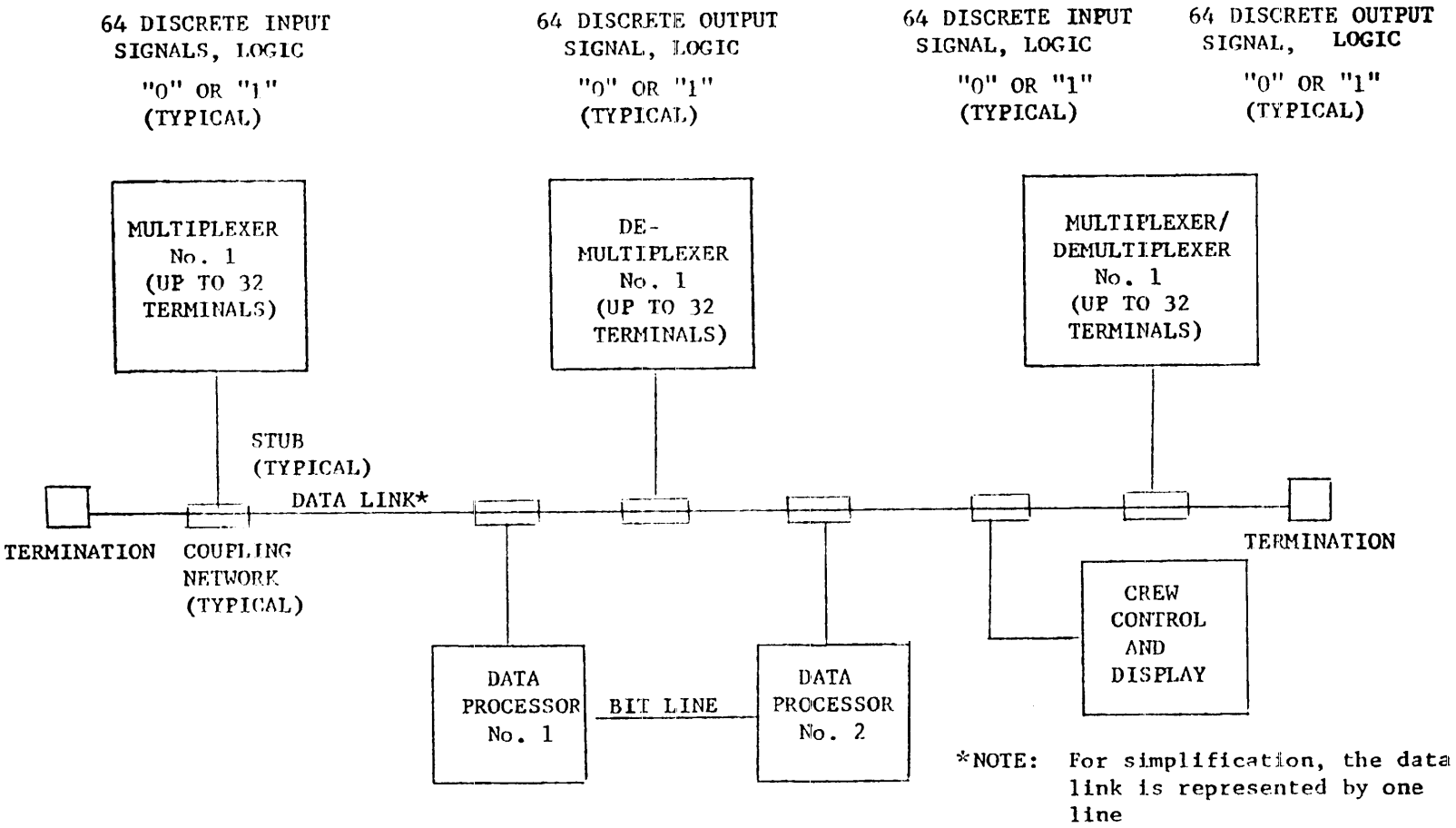


FIGURE 1. Control Group, Electric Power

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2. APPLICABLE DOCUMENTS

2.1 General. The following documents of the issue in effect on the date of invitation for bid form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-B-5087	Bonding, Electrical, and Lightning Protection for Aerospace Systems
MIL-W-5088	Wiring, Aircraft; Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-6781	Control Panel: Aircraft Equipment, Rack or Console Mounted
MIL-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures, Preproduction and Acceptance for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-23103	Thermal Performance Evaluation, Airborne Electronic Equipment; General Requirement for
MIL-I-81219	Indicator, Elapsed Time, Electrochemical

Naval Air Systems Command

AR-5	Microelectronic Devices Used in Avionics Equipment, Procedures for Selection and Approval of
AR-10	Maintainability of Avionics Equipment and Systems, General Requirements for
AR-34	Failure Classification for Reliability Testing, General Requirements for

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AR-104 Reliability Development Test for Avionic Equipment

STANDARDS

Military

MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-415	Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-442	Aerospace Telemetry Standards
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-756	Reliability Prediction Standard
MIL-STD-781	Reliability Tests, Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-1553	Aircraft Internal Time Division Command/Response Multiplex Data Bus

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail Specification. The individual Control Group requirements shall be as specified herein and in accordance with the applicable detail specification. In the event of conflict between requirements of this specification and the detail specification, the latter shall govern.

3.2 Preproduction. This specification makes provision for preproduction testing.

3.3 Parts and Materials. In the selection of parts and material, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Microelectronic technology and microelectronic items shall conform to the requirements specified herein.

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(2) Other parts and materials requirements shall conform to MIL-E-5400.

(3) When previously produced models of this equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.3.1 Nonstandard Parts and Materials Approval. Approval for the use of nonstandard parts and materials (including electron tubes, transistors and diodes) other than microelectronic devices shall be obtained as outlined in MIL-E-5400. Microelectronic devices shall be approved as outlined in AR-5.

3.3.2 Microelectronic Modular Assemblies. Microelectronic modular assemblies shall meet the requirements of AR-5. Conformal coatings, encapsulants, embedments or potting materials used with modular assemblies containing integrated circuits and discrete parts shall be easily removable without damage to the assembly.

3.3.3 Modules. The electronic portions of the equipment shall be functionally modularized in accordance with AR-10.

3.4 Design and Construction. The equipment shall conform with all the applicable requirements of MIL-E-5400 for the design, construction, and workmanship, except as otherwise specified herein, or in the detail specification.

3.4.1 Weight. The weight of each LRU (Line Replaceable Unit) shall be a minimum consistent with good design and shall not exceed the value specified in the detail specification.

3.4.2 Reliability. The contractor shall conduct a reliability program using MIL-STD-785 as a guide. On a reorder from a supplier who has previously produced the equipment, the program previously used may be continued unless otherwise indicated in the contract or order.

3.4.2.1 Operational Stability. The equipment shall operate as specified herein in both the continuous and intermittent mode for at least the minimum period specified in the detail specification, without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.4.2.2 Operating Life. The control group shall have a total operating life as specified in the detail specification. No parts shall require scheduled replacement.

3.4.2.3 Reliability in Mean Time Between Failures. The control group,

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including any built-in-test provisions, shall have an MTBF (Mean Time Between Failures) as specified in the detail specification when tested and accepted as outlined under the requirements of 4.4. The failure of any portion of a redundant LRU shall be considered a control group failure for the purpose of calculating MTBF.

3.4.2.4 Probability of Mission Success. The control group, including any built-in-test provisions, shall have a computed PMS (Probability of Mission Success) as specified in the detail specification when based upon the number of LRUs (Line Replaceable Units) in the group and the mission time specified. Mission success is defined as all outputs being in accordance with discrete bilevel inputs and specified logic.

3.4.2.5 Failure Affecting All Functions. There shall be no single failure which can cause the entire control group to become inoperative.

3.4.2.6 Elapsed Time Recorder. Each LRU shall have provision to record the total accumulated time that electrical power is applied to the LRU. The recorder shall meet the requirements of MIL-M-7793 or MIL-I-81219 as applicable. The recorder shall have a time scale equal to 125% of the LRU operating life (3.4.2.2).

3.4.2.7 Overload Protection. Current overload protection shall be provided for all equipment output wiring. Protection shall be sufficient to prevent damage to aircraft wiring due to overcurrents. Fuzes shall not be utilized.

3.4.3 Cables and Connectors. The equipment shall provide for the use of cables and connectors in accordance with MIL-E-5400.

3.4.3.1 Interconnecting Cabling. The equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of MIL-W-5088. External cables and the connectors attached to the cables shall not be supplied as part of the equipment.

3.4.3.2 Programming Connectors. Connectors used for programming the memory shall be provided with protective environmental captive covers.

3.4.4 Interchangeability. The equipment shall meet the interchangeability requirements of MIL-E-5400.

3.4.5 Provisions for Maintainability. The maintainability program built-in-test features, construction and packaging, provisions for test points, and other maintainability parameters shall be as specified in AR-10. BIT (Built-In-Test) features shall meet the requirements of 3.5.9.

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3.4.5.1 Test Points. Test points shall be provided for safe and convenient access to electronic circuits for all levels of maintenance. The test points shall be compatible with the test equipment designated by the procuring activity for use with the system and each test point shall be protected against inadvertent shorts to ground. The equipment shall be designed to meet the requirements of MIL-STD-415 and AR-10.

3.4.6 Nomenclature, Nameplates, and Identification Marking. Nomenclature and serial number assignments, nameplate approval and identification marking shall be in accordance with MIL-N-18307.

3.4.7 Service Conditions. The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in MIL-E-5400 for Class 3 equipment, except as modified herein.

3.4.7.1 Vibration. The equipment shall operate satisfactorily when subjected to the vibration requirements specified in the detail specification.

3.4.7.2 Explosive Decompression. The equipment shall not be damaged and shall operate as specified herein during and after exposure to an explosive decompression of the surrounding air as detailed in 4.2.2.1.

3.4.7.3 Nuclear Environment. The control group shall withstand the nuclear and electromagnetic pulse environment specified in the detail specification. Acceptable response to the defined environment shall be according to one of the basic categories defined below:

Category I Environment

(a) No permanent degradation of performance shall occur. The transient response shall not trigger any irreversible actuations (e.g., logic circuit change).

(b) The system shall recover to normal performance within the time limits established by system function requirements.

Category II Environment

(a) Output commands shall not change state due to action of the radiation pulse.

(b) The system shall resume normal operation after the pulse.

3.4.7.4 Electromagnetic Compatibility. The control group shall comply with the EMI (Electromagnetic Interference) requirements of MIL-STD-461 (Methods CE01, CE02, CE03, CE04, CS01, CS02, CS06, RE02, RS01, RS02, and

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RS03) except that the test limits of RS03 shall be 10V/M between 14K Hz to 35M and 5V/M between 35M Hz to 10G Hz. The EMC (Electromagnetic Compatibility) control plan shall specify tolerable conducted EMI levels for interconnecting signal lines for procuring activity approval.

3.4.8 Input Electric Power

3.4.8.1 Input Power. The control group shall be designed to accept ac or dc power per MIL-STD-704, Category C. Each LRU shall be capable of accepting power from two separate power sources. The LRU shall function properly with either one or both input power circuits energized. The power drawn from the sources shall be as specified in the detail specification. The two internal power supply circuits shall be electrically isolated within the LRU so that no single power supply failure shall adversely affect the other power supply. Grounding of one power supply shall not impair LRU performance.

3.4.8.2 Power Dissipation. When tested in accordance with 4.2.2.8, the power dissipation for each LRU shall not exceed the values specified in the detail specification.

3.4.9 Circuit Isolation. When tested in accordance with 4.2.2.9, each LRU shall comply with the following circuit isolation requirements:

(a) The LRU data link interface shall be isolated from ground by an impedance as specified in the detail specification.

(b) Isolation of input and output signals from ground shall be as specified in the detail specification.

(c) Each power input shall be isolated from other power inputs and from ground by an impedance as specified in the detail specification.

(d) The input signal channels of the multiplexer and the output channels of the demultiplexer shall be isolated from each other such that a change of logic state, grounding or shorting to supply voltage of one channel shall not affect any other channel of the LRU.

3.4.10 Bonding and Grounding. The equipment shall meet the bonding and grounding requirements of MIL-B-5087. In addition to the case grounds, the plug-in equipment shall also have the case ground established through contacts in the connector. When tested in accordance with 4.2.2.4, the calculated resistance shall not exceed the values given in Table II. The shield for each twisted pair of the data link

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shall be carried through separately and grounded at only one point at the primary processor.

3.4.11 Insulation Resistance. When tested in accordance with 4.2.2.5, the insulation resistance of each LRU shall be as specified in the detail specification.

3.4.12 Dielectric Strength. When tested in accordance with 4.2.2.6, each LRU shall withstand the specified voltage without damage, and there shall be no leakage current in excess of 1 milliampere (ma), nor evidence of damage due to arcing (air discharge), flashover (surface discharge) or breakdown (puncture discharge).

3.4.13 Equipment Cooling. All equipment specified herein shall be capable of operation over the maximum operating temperature-altitude limits specified in MIL-E-5400 for Class 3 equipment with the cooling specified in the detail specification. Thermal design evaluation shall be conducted in accordance with MIL-T-23103.

3.4.14 Warm-Up Time. No warm-up prior to operation shall be required.

3.5 Performance.

3.5.1 Operation. The primary function of the control group shall be to gather discrete control data for electrical power management, solve power management (Boolean) equations utilizing the discrete data as parameters and distribute discrete data to control electrical power functions. The control set shall transfer discrete data using time division multiplexing. The multiplexer and multiplexer/demultiplexer terminals shall accept control signals from aircraft subsystems and equipment and shall condition and convert the signals into serial digital format for baseband transmission on the data link. The demultiplexer and multiplexer/demultiplexer terminals shall receive signals via the data link, decode and condition the signals, and output the signals to the designated loads in their assigned area.

Control group data transfer shall be controlled by one data processor that shall generate address signals which are transferred on the data link to instruct the terminals to transmit or receive groups of data. Dc and low-frequency isolation shall be provided between the data link and the control set LRUs. The data processor shall have logic processing capabilities sufficient to perform logic equation solution, time delays, interlocks, and load management functions involving the discrete signals associated with the aircraft subsystem electric power distributions and control. The data processor logic equation solutions, time delay, interlock and load management functions shall be controlled by a programmable nondestructive read only memory to provide control flexibility. The control group shall have automatic BIT (Built-In-Test) which shall continuously monitor its operation and shall detect and report equipment failures to the LRU level. The BIT shall be controlled by the data processor.

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3.5.1.1 Start Up. Start up is defined as that process which occurs when supply voltage as specified (3.4.8) is applied to each LRU after having experienced supply voltage lower than Locus 3b (Figures 2A and 2B). At start up, all control set LRU output signals shall come on in the "power off" (logic zero) state in such a manner that no incorrect transmission occurs.

3.5.1.2 Normal Operation. The control group shall perform as specified (3.5.1) with supply voltage between Locus 1 and 6 of Figure 2A for ac or Figure 2B for dc applied and LRU start up completed.

3.5.1.3 Through-Put Time. During normal operation, through-put time shall be the maximum elapsed time from change of a discrete input signal of a multiplexer to the resulting output signal change of a demultiplexer with no programmed time delays and with no subcommutation or supercommutation used. Through-put time shall be as specified in the detail specification.

3.5.1.4 Supply Voltage Interruptions. When supply voltage (3.4.8) falls to a value between Locus 6 and 3A of Figure 2A for ac and Figure 2B for dc and returns to a value above Locus 6, each LRU shall return to normal operation.

3.5.1.5 Shut Down. Shut down is defined as that process which occurs when the supply voltage (3.4.8) falls below Locus 3B of Figure 2A for ac and Figure 2B for dc. At shutdown, all control group LRU output signals shall go to the "Power off" (logic zero) state, in such a manner that no incorrect transmission occurs.

3.5.2 Data Link. The system shall include a data link to provide a communication path for information transfer between the data processor(s) and the remote terminals. This information will include system control, LRU status and power management data. The data link shall have separately routed independent redundant paths. Each path shall be accessible to the primary and redundant sections of the data processor(s) and remote terminals. Data link faults on any path shall not affect operation of the other path(s). The level of redundancy and mode of transmission shall be as specified in the detail specification.

3.5.2.1 Transmission Medium. The transmission medium shall consist of transmission cables, stub cables, and coupling networks and termination networks. The transmission medium shall be operated in an electrically balanced condition. Dc and low frequency ac (up to 3k Hz) isolation shall be provided between the transmission cable(s) and the stubs and between the stub cables and the LRUs of the control group. The isolation shall meet the requirements of 3.4.9. Transmission medium characteristics shall meet the requirements of MIL-STD-1553.

3.5.2.1.1 Transmission and Stub Cables. The data link cables shall be shielded, jacketed, uniformly twisted pairs with a controlled impedance and distributed capacitance as specified in the detail specification. The maximum length of these cables shall be as specified. The combined

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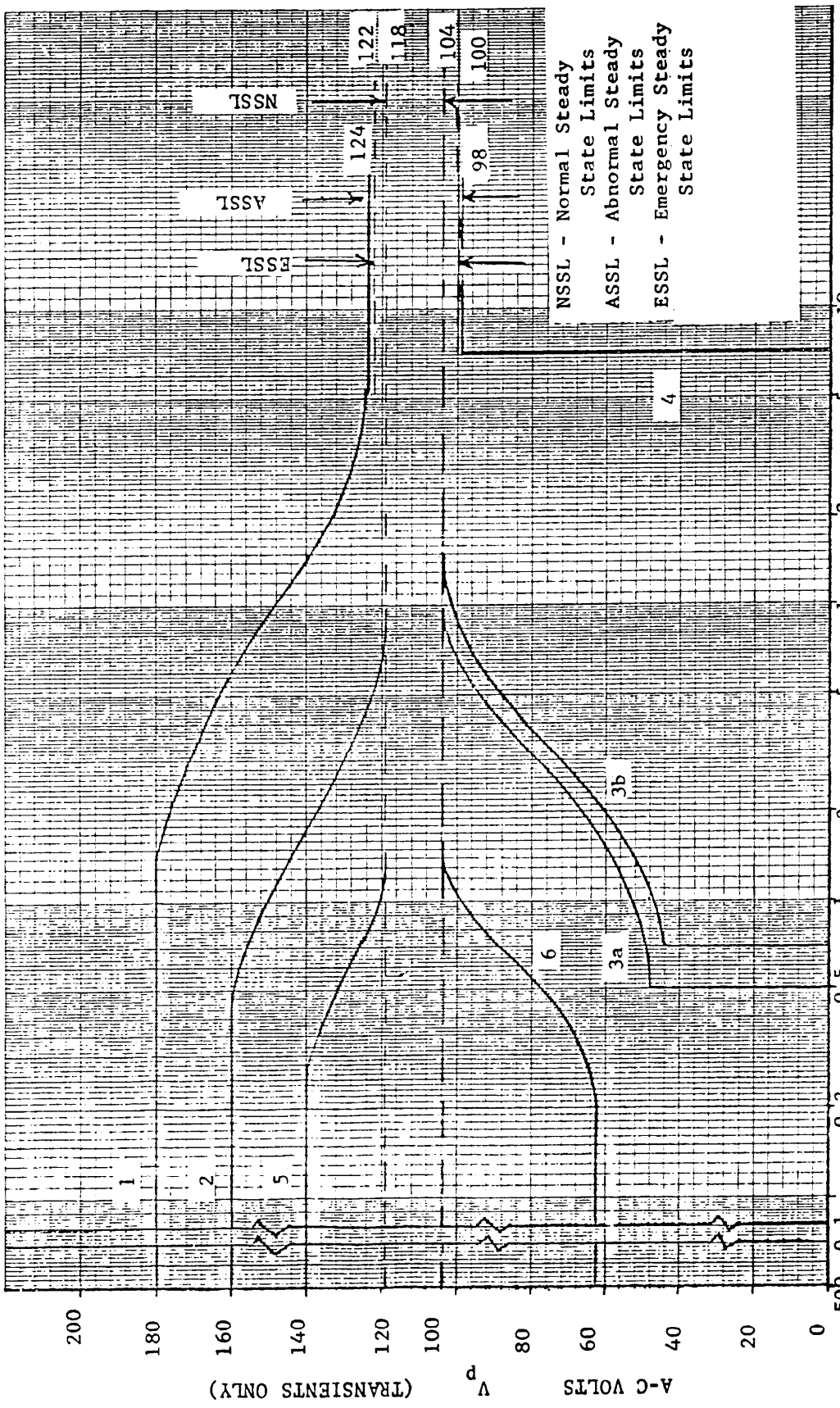


FIGURE 2A TRANSIENT SURGE AC VOLTAGE STEP FUNCTION LOCI LIMITS FOR CATEGORY C EQUIPMENT

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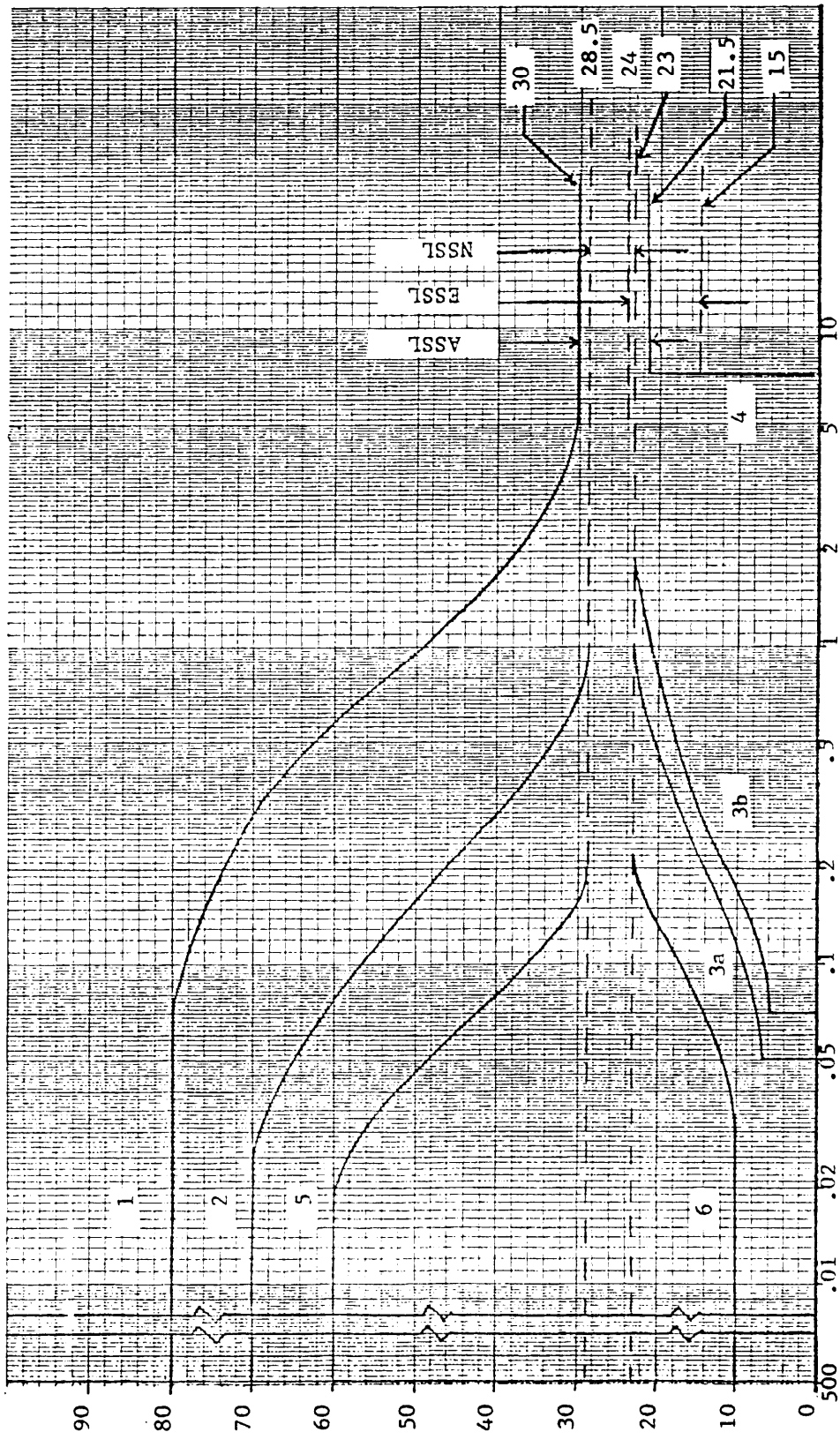


FIGURE 2B TRANSIENT SURGE DC VOLTAGE STEP FUNCTION LOCI LIMITS FOR CATEGORY C EQUIPMENT

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length of the transmission cable and all stubs shall not exceed 300 feet unless otherwise specified.

3.5.2.2 Modulation. The information signal shall be transferred over the data link in serial pulse code modulated form, utilizing biphase-level (Manchester II + 180°) modulation as defined in MIL-STD-442.

3.5.2.3 Voltage Level. The output voltage of each LRU line driver when operating into a load equal to the cable characteristic impedance shall be as specified in the detail specification. The LRU receiver threshold voltage shall be as specified.

3.5.2.4 Bit Rate. The data link bit rate shall be one megabit per second ± 0.01%. When measured in accordance with 4.2.2.1, the rise and fall time of the signal at the terminals of the LRU line driver shall be as specified in the detail specification.

3.5.2.5 Undetected Bit Error Rate. For data transfer between units of the control group, the undetected bit error rate shall be as specified in the detail specification.

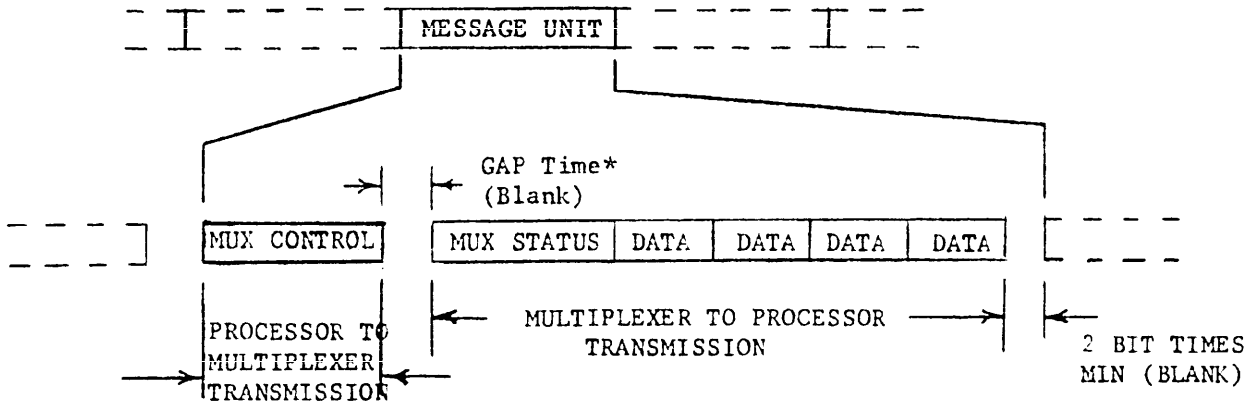
3.5.2.6 Message Format. Information on the data link shall be configured into message units. Each message unit shall consist of the control and status words and one to four words as shown in Figure 3A. Data transmission between the remote terminals and the data processor is initiated by the data processor which generates control words as required.

3.5.2.6.1 Word Format. Each word shall be 20 bit times in length and contain the information and bit assignment as shown in Figures 3B and 3C.

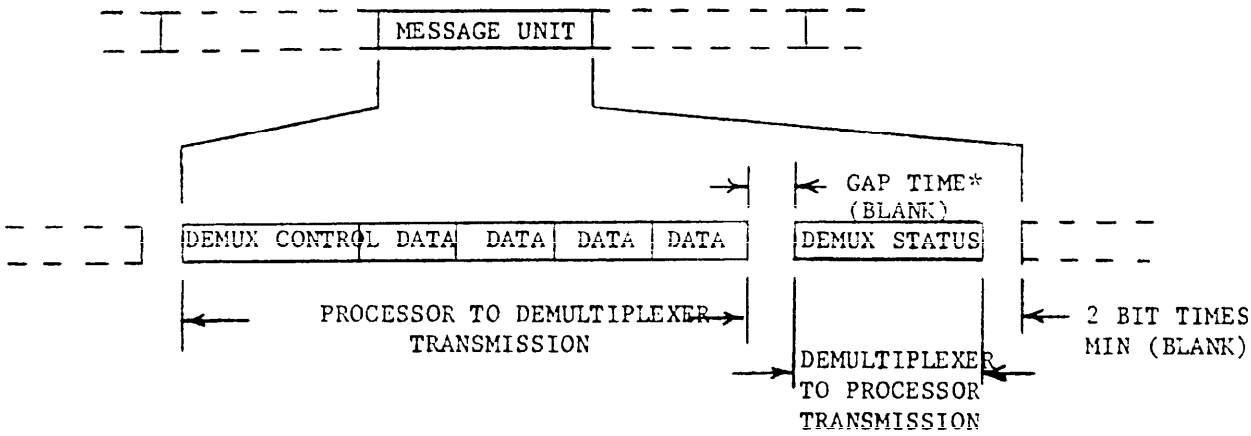
3.5.2.6.1.1 Sync Pattern. The sync pattern for each word shall be an invalid Manchester code of three bits (bit 1, 2, and 3) in duration. For the control and status words, the sync pattern shall be a positive signal for one and one-half bit times followed by a negative signal of one and one-half bit times. For the data words, the sync pattern shall be a negative signal of one and one-half bit times followed by a positive signal of one and one-half bit times.

3.5.2.6.1.2 Terminal Address. The terminal address shall establish the terminal identification. In the control word, bits 4 through 8 shall be a terminal address code generated by the processor. This terminal address code shall be used by the terminals to identify which terminal is being addressed. In the status words, bits 4 through 8 shall contain the terminal address code for that particular terminal. The terminal address code shall be established by a hardware code using the connector pins of the line connector(s) to the terminal as specified in the detail specification. The terminal address code in the status word shall be used by the processor to identify the responding terminal as a check to confirm that the correct terminal has responded.

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PROCESSOR - MULTIPLEXER TERMINAL MESSAGE UNIT

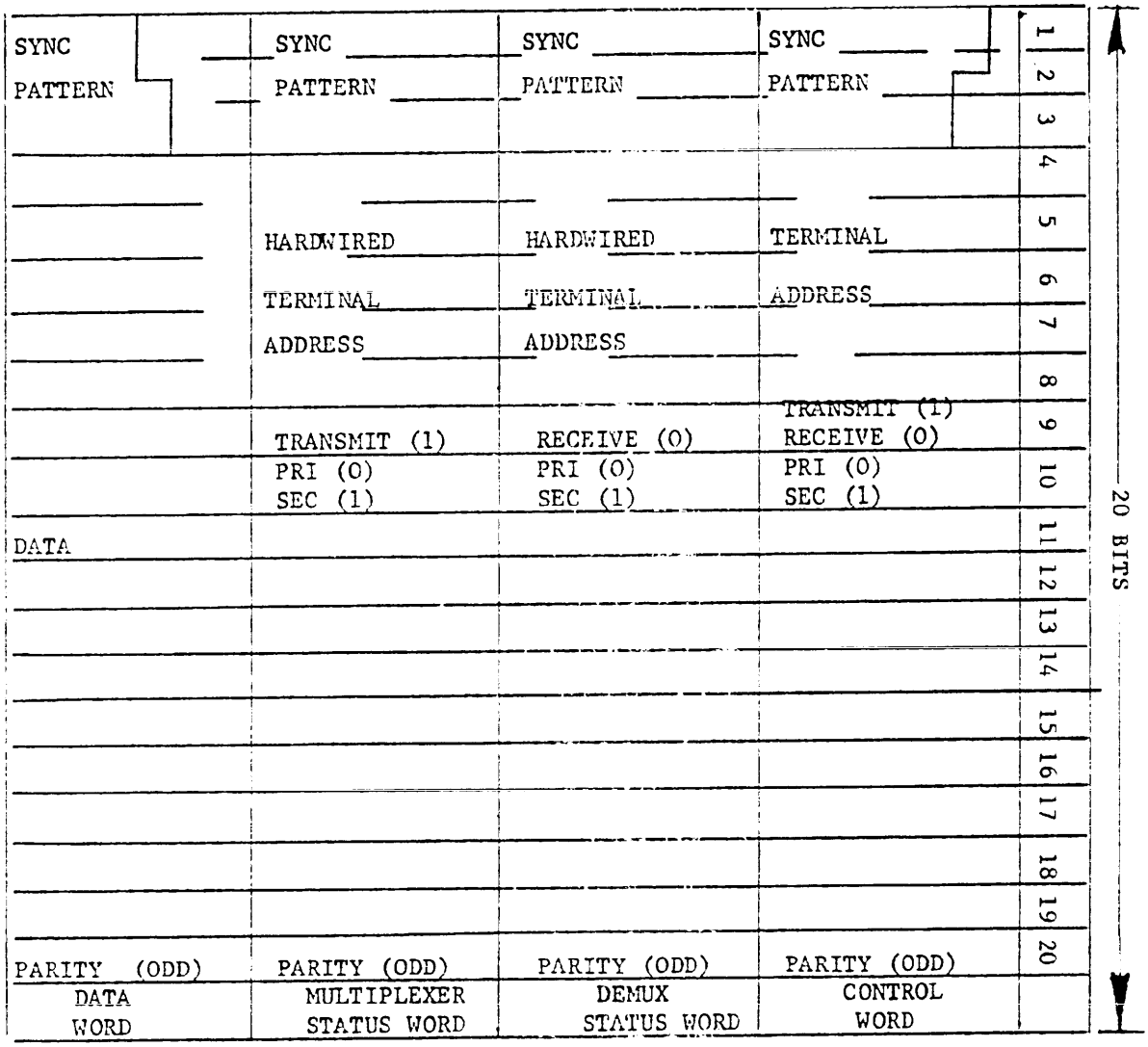


PROCESSOR - DEMULTIPLEXER TERMINAL MESSAGE UNIT

* GAP Time as specified

FIGURE 3A MESSAGE FORMATS

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NOTE: Sync pattern for data word is opposite to pattern for control word.

FIGURE 3B WORD FORMAT

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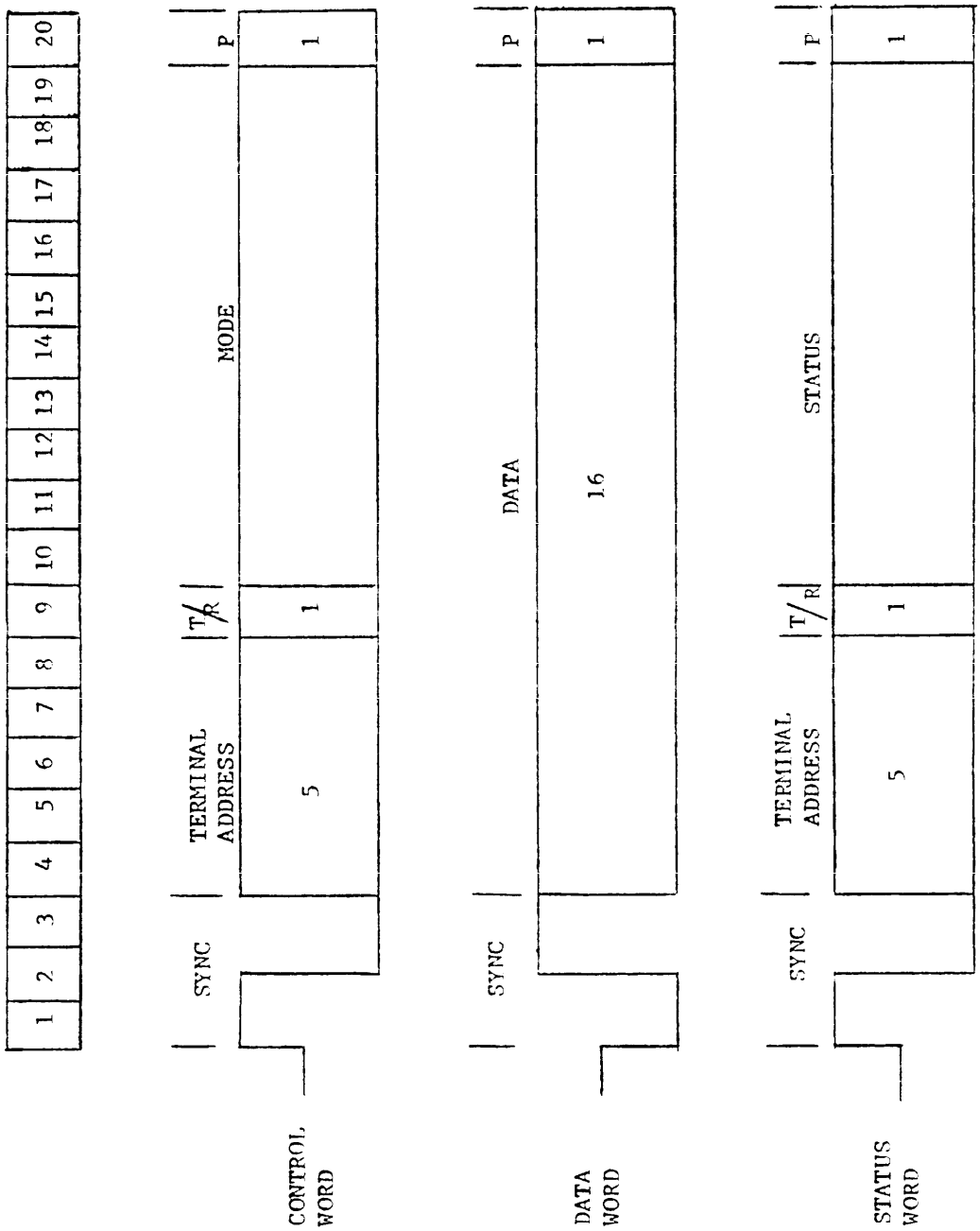


FIGURE 3C WORD FORMAT

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3.5.2.6.1.3 Transmit/Receive Command. Bit 9 shall be the transmit/receive command bit. Bit 9 of the control word shall be used to command the terminals to transmit or receive data. Bit 9 of the status word shall repeat to the processor the transmit/receive command of the control words. A logical "1" in bit 9 shall represent a transmit command and a logical "0" shall represent a receive command.

3.5.2.6.1.4 Mode Code. The mode code in the control word establishes the mode of operation; i.e., when redundance is provided, an LRU may contain dual redundant circuitry with one set operating as the primary set while the other set operates as the secondary (stand-by) set. The processor will normally address the primary set but will address the secondary set after a failure of the primary set is detected by BIT. Bit number 10 is assigned a logic one or zero for this purpose as shown in Figure 3B. Other modes of operation are identified by bit numbers 11 through 19 as specified in the detail specification.

3.5.2.6.1.5 Status Code. The status code in the status word shall be bits numbers 11 through 19 as specified in the detail specification.

3.5.2.6.1.6 Parity. The last bit (bit 20) of each word shall be an odd parity bit. The parity bit shall be generated from preceding bit positions numbers 4 through 19.

3.5.2.6.1.7 Data Field. The data word data field shall be 16 bits. These bits shall include bits 4 through 19.

3.5.3 Data Processor. The data processor shall consist of either a general purpose digital computer plus an appropriate I/O (Input/Output) or a special purpose processor organized to provide the interface and computational requirements specified herein. The computer shall have a parallel arithmetic and logic unit and memory system with a minimum of 16 bits per word for storing the instructions of the application programs, partial results and input/output data. The memory system shall be organized such that instructions are stored in a nondestructive readout, nonvolatile medium that can be altered only through the use of ground support equipment. Partial results and input-output data shall be stored in a scratch pad memory alterable under processor control. The maximum number of memory words required to perform the application programs (3.5.10.1) shall not exceed seventy-five percent (75%) of the memory system supplied such that twenty-five percent (25%) of the instruction and scratch pad memories shall be reserved. No single element failure shall cause complete loss of either instruction or scratch pad memory function.

3.5.3.1 Function. The data processor shall execute the applications software (3.5.10.1), gather data from remote terminals, solve Boolean equations utilizing these data, and distribute the equation solutions to the remote terminals. It shall also provide the signals needed to control the transfer of data within the control group. The data processor shall include its own internal source of timing, and shall synchronize the operation of all units of the control group in both normal and test modes

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of operation. In addition, it shall assign load management priorities to the vehicle electrical loads, and shall perform self test.

3.5.3.2 Performance.

3.5.3.2.1 Start Up. At start up (3.5.1.1), the processor shall automatically execute self test programs, which includes monitoring of multiplexer and demultiplexer BIT, then sample multiplexer data and solve Boolean equations as programmed before outputting any instructions to demultiplexers. This procedure shall take place within the time specified in the detail specification.

3.5.3.2.2 Operation. The data processor shall provide control of all data transfer, solution of control (Boolean) equations, load management functions, and built-in-test.

3.5.3.2.2.1 Data Transfer Control. The data processor shall interface with the data link and thereby control all data transmission and reception by the remote terminals. The commands transmitted to perform this control shall contain synchronization information, identify the terminal addressed, and contain sufficient data to enable the terminal to determine the action it should take in response.

3.5.3.2.2.1.1 Input/Output Memory. The data processor shall have a bit addressable random access read-write memory for storage of input data from data terminals and output data for transmission to data terminals. The capacity of this memory shall be as specified in the detail specification.

3.5.3.2.2.2 Boolean Processing. The data processor shall be capable of solving Boolean equations, utilizing as parameters the input data from the multiplex terminals. The data processor shall contain all necessary computational circuitry, and shall include an instruction memory in which the instructions for solution are stored. The data processor shall be capable of solving combinational or sequential Boolean equations in the numbers, types, and complexities specified in the detail specification.

3.5.3.2.2.3 Load Management. The data processor shall be capable of performing the air vehicle load management which consists of determining air vehicle mode or status from certain data inputs, and of energizing or de-energizing electrical functions to achieve modified system performance in response to mode or status changes. The priorities assigned to various electrical functions in this process shall be programmable.

3.5.3.2.2.4 Built-In-Test. The data processor shall incorporate redundancy and built-in-test features as required such that reliability and performance goals can be achieved. The data processor shall supply all commands and other signals necessary to control and synchronize the operation of built-in-test functions in other units of the control group.

3.5.3.2.3 Supply Voltage Interruption. The processor shall pause and maintain memory for the duration of the interruption (3.5.1.4).

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3.5.3.2.4 Shut Down. When shut down (3.5.1.5) occurs, the processor shall cease operation in such a manner that no incorrect transmissions occur.

3.5.3.3 Interface Characteristics. The data processor shall interface with redundant power sources as defined in 3.4.8 and with redundant data paths as defined in 3.5.2 as specified in the detail specification.

3.5.3.4 Reliability. The MTBF (Mean-Time-Between-Failure) of the data processor shall be as specified in the detail specification. The MTBF shall be demonstrated in accordance with 4.4.

3.5.3.5 Elapsed Time Recorder. The processor shall have provisions to record the total accumulated time that electrical power is applied to the processor. Display of the total elapsed time shall be as specified in the detail specification.

3.5.4 Crew Control Panel. The crew control panel shall interface with the data link to communicate with the processor, provide a display of system status, and initiate control logic functions. To the extent practicable, the crew control panel shall conform to the applicable requirements of MIL-C-6781. The configuration of the panel must be approved by the procuring activity prior to preproduction testing.

3.5.4.1 Function. The crew control panel shall provide the following capabilities:

(a) Display of power controllers which have been tripped due to overload.

(b) Command the processor to reset a tripped power controller.

(c) Select and open any power controller.

(d) Override group built-in-test to allow selection of the primary or secondary block function of any individual unit or the entire group.

(e) Display faults in individual units (LRUs) which have caused block switching or unit functional failure.

3.5.4.2 Performance.

3.5.4.2.1 Start Up. At start up (3.5.1.1), the panel shall become operational within the time specified in the detail specification such that no incorrect transmission occurs and all displays are cleared.

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3.5.4.2.2 Operation. The panel display data shall be compiled by the processor and transmitted via the data link using the data format of Figures 3A and 3B and as specified in the detail specification. After recognition of the proper address by the panel, data words shall be received and decoded and the information displayed as specified. Requests for data initiated from the panel shall be coded in data words and transmitted to the processor per Figures 3A and 3B. Upon receipt of these commands, the processor logic shall transmit data to the appropriate units (LRUs) to carry out the action.

3.5.4.2.3 Supply Voltage Interruption. The panel shall retain its data for the duration of the interruption (3.5.1.4), such that at the conclusion of the interruption the outputs are immediately returned to the pre-interruption state.

3.5.4.2.4 Shut Down. When shut down (3.5.1.5) occurs, the panel shall cease operation in such a manner that no incorrect transmissions occur.

3.5.4.2.5 Redundancy and Built-In-Test. The panel shall incorporate redundancy in power supplies and data link interfaces. The redundant elements shall be automatically selected by built-in-fault detection circuits. The operating status of the panel, including the off line redundant elements, shall be monitored continuously and reported to the processor in the status word.

A manually initiated press to test capability shall be incorporated for all display elements.

3.5.4.3 Interface Characteristics. The panel shall interface with redundant power sources as defined in 3.4.8 and with redundant data paths as defined in 3.5.2 as specified in the detail specification.

3.5.4.4 Reliability. The MTBF for the crew control panel shall be as specified in the detail specification. The MTBF shall be demonstrated in accordance with 4.4.

3.5.4.5 Elapsed Time Recorder. The crew control panel shall have provisions to record the total accumulated time that electrical power is applied to the crew control panel. Display of the total elapsed time shall be as specified in the detail specification.

3.5.5 Multiplexer.

3.5.5.1 Function. The multiplexer shall interface with the data link and with 64 discrete bilevel signal inputs. It shall be capable of receiving control words from the data link, determining the validity of received control words and recognizing its own unique address within the control words. In response to a valid control word containing its unique address, the multiplexer shall respond by sampling, multiplexing, and transmitting the state of some or all of its inputs as instructed by the control words.

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3.5.5.2 Performance.

3.5.5.2.1 Start Up. At start up (3.5.1.1), the multiplexer shall become operational within the time specified in the detail specification such that no incorrect transmission occurs.

3.5.5.2.2 Operation. The multiplexer shall receive and decode control words from the data link and, on reception of a valid control word containing the unique address of the multiplexer, shall sample, encode, and transmit in numerical order, the state of some or all of its 64 bilevel inputs as instructed by the control word. If disturbances which could affect data accuracy are detected, the multiplexer shall not transmit data during the disturbance. If a transmission is in progress at the time of a disturbance which could affect data accuracy, the transmission shall be terminated and shall not be reinstated until receipt of a valid control word.

3.5.5.2.3 Supply Voltage Interruptions. During supply voltage interruption (3.5.1.4), the multiplexer shall either continue normal operation or follow a shut down, start up sequence.

3.5.5.2.4 Shut Down. When shut down (3.5.1.5) occurs, the multiplexer shall cease operating in such a manner that no incorrect data transmission occurs.

3.5.5.2.5 Redundancy and Built-In-Test. The multiplexer shall incorporate redundancy and built-in-test provisions as necessary such that reliability and performance goals can be achieved.

3.5.5.3 Interface Characteristics. The multiplexer shall receive instructions from and transmit data onto the data link as specified in the detail specification. No single multiplexer failure shall adversely affect the operation of the data link. The multiplexer shall accept external electrical power as specified (3.4.8.1).

3.5.5.3.1 Data Input Signal Characteristics. The multiplexer shall accept 64 discrete bilevel input signals with characteristics as specified in the detail specification.

3.5.5.4 Reliability. The MTBF of the multiplexer shall be as specified in the detail specification. The MTBF shall be demonstrated in accordance with 4.4.

3.5.5.5 Elapsed Time Recorder. The multiplexer shall have provisions to record the total accumulated time that electrical power is applied to the multiplexer. Display of the total elapsed time shall be as specified in the detail specification.

3.5.6 Demultiplexer.

3.5.6.1 Function. The demultiplexer shall interface with the data link and provide 64 continuous discrete bilevel outputs. It shall receive control words from the data link, determine the validity of received control words, and recognize its own unique address within a control word. In response to a valid control word, the demultiplexer shall accept one or

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more data words from the data link, check the validity of the data words and demultiplex the data bits of those words to some or all of its outputs as commanded by the control word.

3.5.6.2 Performance.

3.5.6.2.1 Start Up. At start up (3.5.1.1), the demultiplexer shall become operational within the time specified in the detail specification, such that no incorrect transmission occurs. The discrete bilevel outputs shall come on and remain in the "Power Off" (logic zero) state until valid data is received.

3.5.6.2.2 Operation. The demultiplexer shall provide 64 continuous bilevel signal outputs as specified in the detail specification and shall integrate (3.5.6.3.3) and update the output signals when valid control and data words are received from the data link. The demultiplexer shall utilize the data received to control the outputs such that data Bit No. 1 controls output No. 1, data Bit No. 2 controls output No. 2, and so on in numerical order. Those outputs not commanded to change by the data word shall retain their logical states.

3.5.6.2.3 Supply Voltage Interruption. The demultiplexer shall retain its data for the duration of the interruption (3.5.1.4) such that at the conclusion of the interruption, the outputs are immediately returned to the last state prior to the interruption and shall hold that state until commanded to change by the reception of valid control and data words.

3.5.6.2.4 Shut Down. When shut down (3.5.1.5) occurs, the demultiplexer shall cease operation in such a manner that no incorrect signal transmission occurs. The discrete bilevel outputs shall go to the "Power Off" (logic zero) state.

3.5.6.2.5 Redundancy and Built-In-Test. The demultiplexer shall incorporate redundancy and built-in-test provisions as necessary such that specified reliability and performance goals can be achieved.

3.5.6.3 Interface Characteristics. The demultiplexer shall receive instructions and data from and transmit data onto the data link as specified in the detail specification. No single demultiplexer failure shall adversely affect the operation of the data link. The demultiplexer shall accept external power as specified (3.4.8.1). The demultiplexer shall have 64 continuous discrete bilevel outputs.

3.5.6.3.1 Output Driver Signal Characteristics. The demultiplexer shal furnish 64 signal outputs with characteristics as specified in the detail specification.

3.5.6.3.2 Output Integration. Data integration shall be provided at the demultiplexer output so that two (2) successive and identical signals must be received to result in a change at the output.

3.5.6.4 Reliability. The MTBF of the demultiplexer shall be as specified in the detail specification.

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3.5.6.5 Elapsed Time Recorder. The demultiplexer shall have provisions to record the total accumulated time that electrical power is applied to the demultiplexer. Display of the total elapsed time shall be as specified in the detail specification.

3.5.7 Multiplexer/Demultiplexer.

3.5.7.1 Function. The multiplexer/demultiplexer is a remote terminal which performs the function of the multiplexer and the demultiplexer. The multiplexer/demultiplexer interfaces with the data link and 64 continuous discrete bilevel signal inputs, and provides 64 discrete bilevel signal outputs. In response to a valid control word containing its unique address, the multiplexer/demultiplexer shall either transmit digital data on the data link or shall receive and demultiplex data from the data link, as commanded by the control word.

3.5.7.2 Performance.

3.5.7.2.1 Start Up. The provisions of 3.5.5.2.1 and 3.5.6.2.1 shall apply to the start up performance of the multiplexer/demultiplexer.

3.5.7.2.2 Operation. The provisions of 3.5.5.2.2 and 3.5.6.2.2 shall govern the multiplexing and demultiplexing operations, respectively, of the multiplexer/demultiplexer.

3.5.7.2.3 Supply Voltage Interruption. The provisions of 3.5.5.2.3 and 3.5.6.2.3 shall apply to the supply voltage interruption of the multiplexer/demultiplexer.

3.5.7.2.4 Shut Down. The provisions of 3.5.5.2.4 and 3.5.6.2.4 shall apply to the shut down performance of the multiplexer/demultiplexer.

3.5.7.2.5 Redundancy and Built-In-Test. The multiplexer/demultiplexer shall incorporate redundancy and built-in-test provisions as necessary such that reliability and performance goals can be achieved.

3.5.7.3 Interface Characteristics. The provisions of 3.5.5.3.2 and 3.5.6.3.2 shall apply.

3.5.7.3.1 Data Input Signal Characteristics. The provisions of 3.5.5.3.1 shall apply.

3.5.7.3.2 Output Driver Signal Characteristics. The provisions of 3.5.6.3.1 shall apply.

3.5.7.3.3 Output Integration. The provisions of 3.5.6.3.3 shall spply.

3.5.7.4 Reliability. The MTBF of the multiplexer/demultiplexer shall be as specified in the detail specification. The MTBF shall be demonstrated in accordance with 4.4.

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3.5.7.5 Elapsed Time Recorder. The multiplexer/demultiplexer shall have provisions to record total accumulated time that electrical power is applied to the multiplexer/demultiplexer. Display of the total elapsed time shall be as specified in the detail specification.

3.5.8 Data Link Coupler/Terminator.

3.5.8.1 Function. The DLC/T (Data Link Coupler/Terminator) shall interface each of the control group LRUs with the data link. The coupling network of the DLC/T shall be as specified in the detail specification and shall be used to connect the stub cables to the main transmission cables. The input/output coupling characteristics, coupling coefficient, and impedance transformations will be such that the imposed mismatch on the main transmission cable shall be less than a voltage standing wave ratio of 2/1 and shall maintain the system bit error rate specified in the detail specification. The termination network of the DLC/T shall be as specified in the detail specification and shall be capable of terminating the transmission cable in its characteristic impedance.

3.5.8.2 Operation. The data link coupler/terminator shall be used to couple the stub cables to the transmission cables and, in addition, terminate the transmission cable in its characteristic impedance. The DLC/T coupling network shall present an impedance to the transmission cable as specified in the detail specification and an impedance to the stub cable equal to the characteristic impedance of the stub cable as specified in the detail specification. The coupling network shall provide sufficient isolation so that grounds, shorts, or loss of any stub cable to any control group LRU will not degrade the transfer of data between all other control group LRUs. The DLC/T termination network shall consist of a passive network which presents an impedance to the transmission cable equal to the nominal characteristic impedance of the transmission cable.

3.5.8.3 Interface Characteristics. The data link coupler/terminator shall interface with the transmission cable and stub cable as specified herein and in the detail specification.

3.5.8.4 Reliability. The MTBF of the data link coupler/terminator shall be as specified. The MTBF shall be demonstrated in accordance with 4.4.

3.5.9 Built-In-Test.

3.5.9.1 General BIT Reporting. The control group shall incorporate specific built-in-test monitoring in its individual LRUs to accomplish redundant block switching and operational status reporting. Each LRU shall report its BIT status to the processor via the data link utilizing the status word format of Figures 3A and 3B. The processor shall also have the capability of forcing each multiplexer and demultiplexer into either redundant state based on the BIT stored program or upon manually initiated commands from the crew control panel (3.5.4). The processor shall correlate the reported BIT data (along with other system status indications such as PC trip) to determine LRU and control group status.

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This processor correlated fault/BIT status shall be reported via the crew display panel (3.5.4) and/or special function GSE (Ground Support Equipment).

3.5.9.2 BIT Performance. The circuits and devices which provide BIT functions shall be designed in such a manner that failures of BIT circuitry do not adversely affect the control group operation. BIT dependability shall meet the requirements of AR-10.

3.5.9.2.1 BIT Test Times. BIT shall perform the specified functions within the following time constraints:

(a) Fault Detection - The BIT fault detection data shall be updated at intervals specified in the detail specification.

(b) Performance Degradation - Performance degradation data shall be available for presentation within two seconds from the time that data is requested.

(c) Fault Isolation - Fault isolation data shall be provided concurrently with fault detection and performance degradation data wherever possible. Initiated fault isolation testing shall be accomplished in one minute or less from the initiation of test.

3.5.10 Software.

3.5.10.1 Application Software. The application software shall consist of all programs resident in the data processor required to meet the functional requirements specified herein. The programs shall include but not be limited to, data transfer control, air vehicle logic control, load management, time delay, interfaces, built-in-test, display and executive. The application software shall be written in an improved high order language as specified in the detail specification.

3.5.10.2 Automatic Test Equipment Software. The ATE (Automatic Test Equipment) software shall consist of all software required to operate the ground support equipment to fault isolate any unit of the control group. The ATE software shall be written as specified in the detail specification.

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3.5.10.3 Support Software. The support software shall consist of all software required to assemble, compile, debug, load, generate system simulation, and test and integrate all application and ATE software. All of the support software shall be hosted on the same computer and shall be written in an approved language as specified in the detail specification. The selection of the host computer system shall be approved by the procuring activity prior to commencement of the programming effort.

3.5.10.4 Software Documentation. All application, ATE and support software shall have both user and design documentation. Schedules for submission of all software documentation shall be approved by the procuring activity prior to commencement of the programming effort. All software documentation shall be submitted to the procuring activity for approval and acceptance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests. Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction (First Article) Tests

(2) Initial Production Tests

(3) Acceptance Tests

(4) Life Tests

4.1.2 Standard Conditions. The following conditions shall be considered standard for establishing normal performance characteristics and for making laboratory bench tests.

Temperature	Room ambient ($25^{\circ}C \pm 10^{\circ}C$)
Altitude	Normal ground (0 - 5000 ft)

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Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 5.0 VAC 400 Hz and/or 27.5 ± 2.0 VDC

4.2 Preproduction (First Article) Tests - Preproduction tests shall be conducted by the contractor on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.6. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) Test Data - The contractor shall submit all data collected in conducting these tests to the procuring activity for review.

4.2.2 Scope of Tests. Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification, other applicable specifications and the contract. Preproduction tests shall include environmental tests in accordance with the procedures of MIL-T-5422, as well as the tests detailed herein. Reliability development test shall be performed in accordance with the requirements of AR-104.

4.2.2.1 Explosive Decompression. The equipment shall be subjected to an explosive decompression of the surrounding air from a level of 11.1 psia to a level of 0.65 psia. The time of decompression shall not exceed 0.01 seconds. The rate of change at any time shall be at least 1045 psi per second. The test sample shall not be damaged by the exposure to the decompression and shall perform as specified during and after the decompression.

4.2.2.2 Electromagnetic Pulse. An EMP (Electromagnetic Pulse) test shall be conducted to verify compliance with the requirements of 3.4.7.3. An EMP test plan shall be submitted to the procuring activity for approval.

4.2.2.3 Electromagnetic Compatibility. An EMC (Electromagnetic Compatibility) test shall be performed to verify compliance with the EMC requirements of 3.4.7.4. An EMC test plan shall be submitted to the procuring activity for approval.

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4.2.2.4 Bonding and Grounding. A bonding and grounding verification test shall be conducted as follows:

The equipment under test shall be connected as shown in Figure 4. Resistance shall be calculated using ohms law - R = E/I. The test currents and maximum allowable resistance shall be in accordance with Table I.

TABLE I

Wire Size	Test Current	Resistance (Ohms)
26	2.0	0.031
24	3.0	0.020
22	5.0	0.015
20	7.5	0.009
18	10.0	0.007

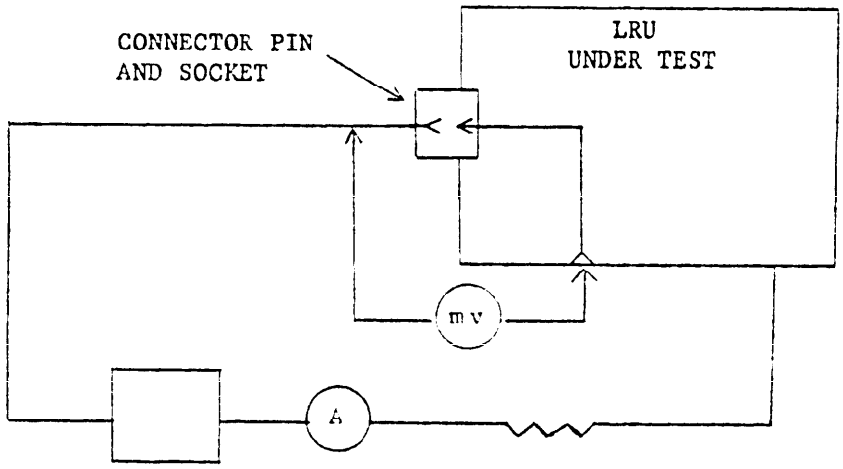


FIGURE 4 - Test Circuit

4.2.2.5 Insulation Resistance. Insulation resistance shall be tested in accordance with Method 302 of MIL-STD-202. The following details shall apply.

(a) The insulation resistance shall be measured between the shorted pins of the connector and the chassis of the LRU.

(b) The insulation resistance shall be measured at ambient (25°C) temperature conditions.

(c) The test voltage shall be 500 volts ± 10%.

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(d) Prior to conducting the insulation resistance and dielectric tests, the equipment shall be examined to determine if any electronic or miniature electromechanical devices are mounted on the frame of the equipment. Since these components are not designed to withstand the test voltages, they may either be removed from the frame of the equipment or individually short circuited. In such cases, a design analysis shall be included with the test results to show that the components in question are adequately protected against transients.

4.2.2.6 Dielectric Strength. LRUs shall be tested as specified in 4.2.2.6.1 and 4.2.2.6.2.

4.2.2.6.1 At Atmospheric Pressure. LRUs shall be tested in accordance with Method 301 of MIL-STD-202. The following details shall apply:

(a) Preparations - Necessary precautions shall be taken to assure that internal components and circuitry are not adversely affected by application of the test voltage.

(b) Test voltage - 1000V RMS.

(c) Nature of potential - AC.

(d) Duration - As specified in MIL-STD-202.

(e) Points of application - The test voltage shall be applied between the shorted pins of the connector and the chassis of the LRU.

(f) Leakage current - 1.0 ma maximum.

(g) Following these tests, LRUs shall be examined for evidence of arcing, flashover, insulation breakdown and damage.

(h) Measurements after test - Insulation resistance.

4.2.2.6.2 At Reduced Barometric Pressure. LRUs specified for operation above 10,000 feet shall be tested in accordance with Method 105 of MIL-STD-202. The following details shall apply"

(a) Method of mounting - Normal mounting means.

(b) Test Condition - D.

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(c) Tests during subjection to reduced pressure - As specified in 4.2.6.7.1 except test voltage shall be 500 volts rms.

(d) Same as 4.2.6.7.1(g).

4.2.2.7 Input Power. With the LRU connected as shown in Figure 5, adjust the power supplies for the input voltage specified in the detail specification. The LRU shall operate as specified with S_1 or S_2 closed; or with S_1 and S_2 closed.

4.2.2.8 Power Dissipation. To perform the power dissipation test the processor must be programmed to set all demultiplexer outputs to the high (1) state for one set of measurements and to low (0) state for another set.

With the control group connected as shown in Figure 6 and all outputs in the high (1) state, measure the current required by each LRU. Calculate the maximum power dissipation according to the relation, P = VI. Repeat procedure with outputs in the low (0) state.

4.2.2.9 Circuit Isolation. Circuit isolation of the data link interface, input and output signals and power supplies from ground and between power supplies shall be determined using the test method illustrated in Figure 7.

Isolation between multiplexer input channels shall be determined as follows:

(a) Set all inputs at "0" logic level. Apply a logic "1" to one channel. Check all other channels for indication of change of state.

(b) Apply a logic "1" to all channels. Change the input to one channel to a logic "0". Monitor all other channels for indication of state.

(c) Repeat the above tests to at least 15 channels randomly chosen.

Isolation between demultiplexer channels shall be determined as follows:

(a) Set all channels to a logic "1" state. Short one channel to ground. Monitor all other channels for indication of a change of state.

(b) Set all channels to a logic "0" state. Apply specified (3.1) control voltage to one channel. Monitor all other channels for indication of change of state.

(c) Repeat the above tests to at least 15 channels randomly chosen.

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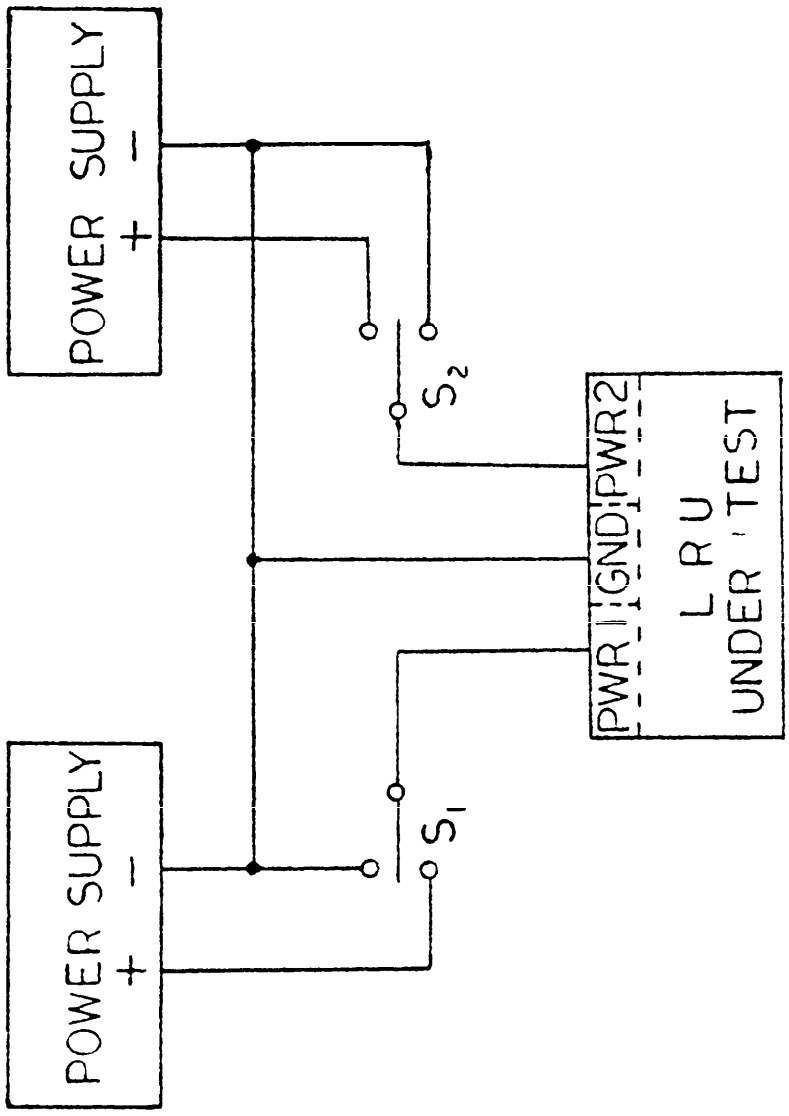


FIGURE 5. Test Circuit

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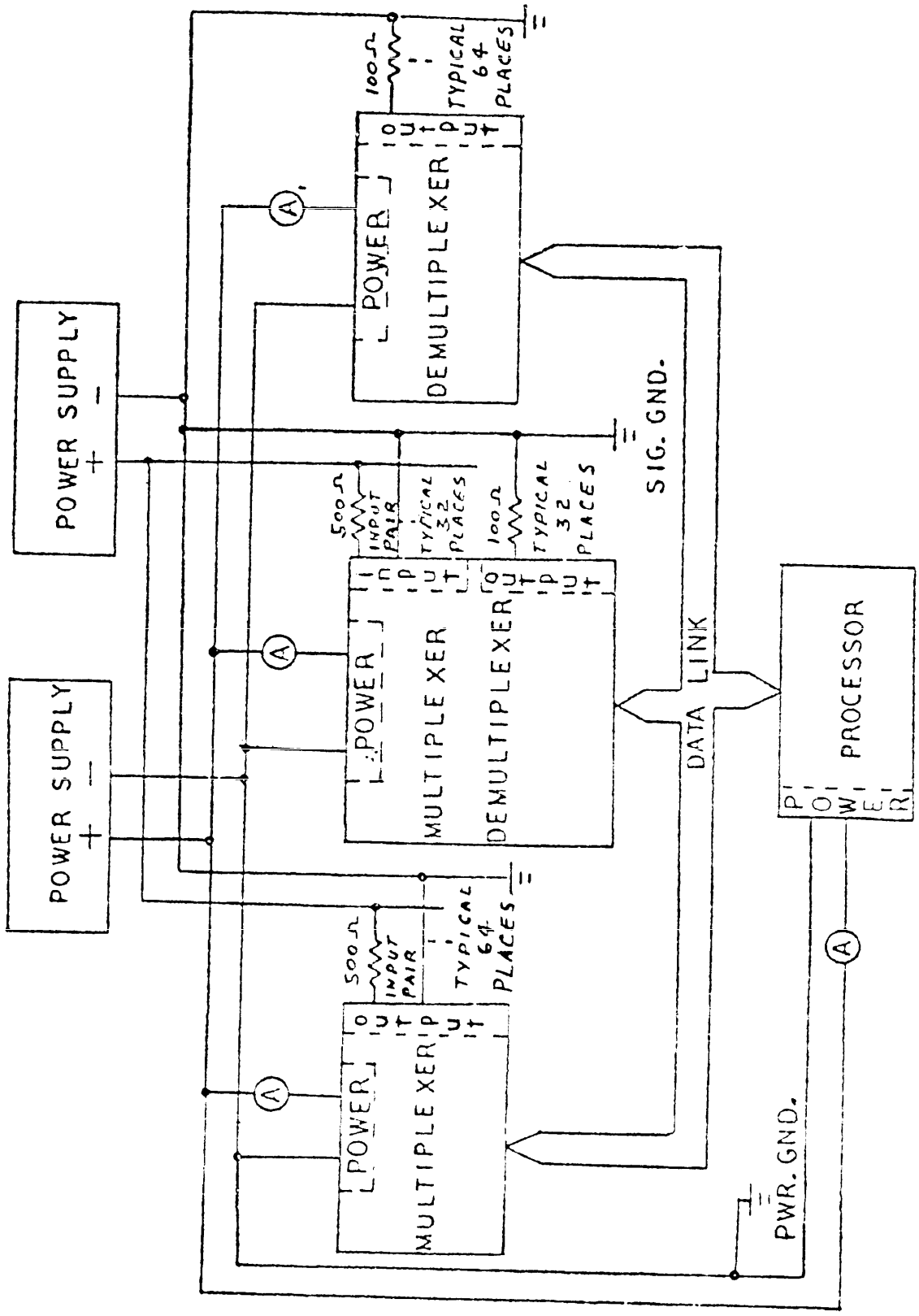


FIGURE 6. Test Circuit

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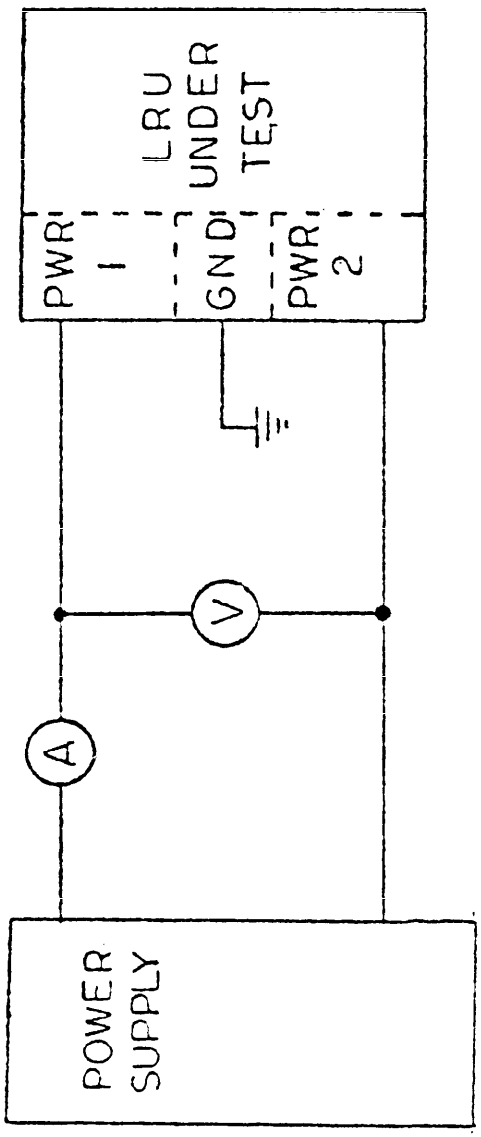


FIGURE 7. Test Circuit (Typical)

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4.2.3 Preproduction (First Article) Approval. Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be delivered as one of the equipments under the contract.

4.2.4 Production Equipments. Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection and for equipment already accepted by the Government. It shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests. One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated Government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests. No other tests shall be conducted on equipment prior to starting the Initial Production Tests. The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests. This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material. In addition to the complete equipment submitted for Initial Production Tests, the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval. Approval of the Initial Production Sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the Initial Production Sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

4.3.4 Reconditioning of Initial Production Test Sample. On completion of the initial production test the equipment shall be reworked by the contractor by replacing all limited life or damaged items. After reworkin the contractor shall resubmit the equipment for acceptance.

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4.4 Acceptance Tests. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests
(2) Sampling Tests
(3) Reliability Assurance Tests
(4) Special Tests

4.4.1 Individual Tests. Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operational Test
(3) Manufacturing Run in Test

4.4.1.1 Examination of Product. Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.4.1.2 Operational Test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run in Test. Each equipment shall be operated under the conditions specified herein for a period of ten hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment. This test shall be deleted if the reliability test includes a test on each equipment which consumes at least ten hours of operation.

Temperature ... Ambient room
Humidity ..... Ambient room
Vibration ..... Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's

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The equipment shall be vibrated (without vibration isolators) for a period of ten minutes prior to the beginning of the ten-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to the plane for five minutes. Where it is not feasible to vibrate the equipment in two directions, the vertical direction shall be used. During the ten-hour period of operation following the ten-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the ten-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The ten-hour period specified above may be composed of two five-hour periods to conform with standard working hours.

4.4.2 Sampling Tests. Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the Government inspector in accordance with the following:

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1#
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

#One out of first ten need not be selected and tested if initial production tests are conducted. Sampling tests are not required when reliability assurance tests are conducted.

4.4.2.1 Scope of Tests. As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environmental conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

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(3) Manufacturing run in test specified in 4.4.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

4.4.3 Reliability Assurance Tests. Reliability assurance tests shall be conducted using MIL-STD-781. Tests are required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted. Classification of failure shall be in accordance with MIL-STD-781 and AR-34.

4.4.3.1 Qualification Phase. Prior to the acceptance of equipments under the contract or order, a minimum of three (3) equipments shall be tested as outlined in MIL-STD-781 under the section entitled "Qualification Phase of Production Reliability Tests." The maximum number of equipments to be used shall be those listed in Table 5 of MIL-STD-781. For the Qualification Phase, Test Level F shall be used. The Accept-Reject Criteria for Test Plan I shall be used.

4.4.3.2 Reliability Production Acceptance (Sampling) Phase Tests. The equipment, throughout production, shall be tested as outlined in MIL-STD-781 (as modified herein) under the section entitled "Production Acceptance (Sampling) Phase of Production Reliability Tests". Test level F shall be used.

4.4.3.2.1 All Equipment Test. Each equipment produced except those submitted for the Reliability Qualification Test, shall be tested for 50 hours. Prior to the 50 hour test on each equipment, a burn-in period may be used at the option of the contractor. If the burn-in period is to be used the details thereof must be included in the approved test procedures. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures thereon (not counting burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan II of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. At the conclusion of each month the test results shall be sent to the procuring activity and to the Naval Air Systems Command, Attention: Avionics Division. At any time that the current totals of test hours and test failures plotted on Test Plan II curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of equipment at any time that a reject situation exists pending a review of the contractor's efforts to improve the equipment, the equipment parts, the equipment workmanship, etc., so that the entire compilation will show other than a reject decision.

4.4.3.3 Test Details. The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, preventive

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maintenance to be allowed during the test, etc., shall be part of the test procedures to be submitted and approved by the procuring activity prior to the beginning of the Qualification Test Phase of the Reliability Assurance Tests.

4.4.4 Special Tests. Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequacy quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule. Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests. Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment Failure. Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test. The contractor shall furnish all samples and shall be responsible for accomplishing the life tests. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual test. The life test shall be performed under the conditions specified in 4.5.1. The life test sample shall be selected

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by the Government inspector in accordance with the following. (Equipments which have successfully passed the Initial Production Test, Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) (Test Life accumulated during the Reliability Test may be counted toward the Life Test provided the entire 300 hours is accumulated on a single equipment and equipment selected is in accordance with the table below.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.1 Test Conditions. The life test shall be conducted under the standard conditions of 4.1.2.

4.5.2 Test Periods. The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check. At approximately eight-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data. The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

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4.5.4.1 Failure Report. In the event of a failure, the Government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Test Procedures. The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment. Equipment which has been subjected to initial production, acceptance and life tests shall be reconditioned by the contractor by replacing all wear or damaged items. After reworking the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing. No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest. Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

5. PREPARATION FOR DELIVERY

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed, and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Ordering Data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

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(a) Title, number, and date of this specification.

(b) Selection of applicable levels of packaging and packing (see 5.1).

6.2 Precedence of Documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract. The contract shall have precedence over any specification.

(b) Specification Sheets. The requirements of the detail specification sheets shall have precedence over this specification.

(c) This Specification. This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(d) Referenced Specifications. Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Performance Objectives. Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigations show advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 Type Designations. The type designation may be modified by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.4.6. The correct type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.5 Abbreviations.

ATE	Automatic Test Equipment
BIT	Built-In-Test

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DLC/T	Data Link Coupler/Terminator
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
EMP	Electromagentic Pulse
GSE	Ground Support Equipment
I/O	Input/Output
LRU	Line Replaceable Unit
MTBF	Mean Time Between Failure
PMS	Probability of Mission Success
SRA	Shop Replaceable Assembly

6.6 Detail Specification Format. The format for preparation of the detail specification is given in Appendix A of this specification

PREPARING ACTIVITY
NAVY-AS
(Project No. 6110-N202)

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-81883(AS) Control Group, Electric Power OX-XXXX(V)/A Gen Spec for

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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